

Are Paris and Lyon Still Monocentric Cities?

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- **Context:** The evolution of major French cities raises questions about their spatial organization.
- **Research Question:** Are Paris and Lyon still monocentric despite urban transformations?
- **Approach:** Analysis of job location since 1968 and economic dynamics.

Key Definitions

- **Monocentric Model** (Alonso, 1964):
 - Concentration of jobs in a **single center**
 - Workers must therefore be **transported** to employment zones
 - Key role of **transport costs** (t) and **decreasing rents** ($L(d)$)
- **Bid-Rent Function:**

$$L(d) = y - t \cdot d - \bar{x}_0$$

Where $L(d)$ = rent at distance d , y = income, t = cost/km, $\bar{x}_0$ = residual consumption

- Households trade off high rent near the center ($d=0$) against transport costs in the periphery.
- **Polycentrism:**
 - Emergence of **secondary centers** (e.g., La Défense)
 - Increase in housing in these centers
 - Improved local transport links (e.g., La Défense - Paris in 10 min)

Model Assumptions and Limitations

The 5 Fundamental Assumptions (Alonso-Muth-Mills)

- 1 Jobs are **centralized** and **concentrated** at a single point
- 2 Households are **identical** in preferences/income
- 3 Housing is **homogeneous**
- 4 Real estate **production**: $L(d) = (r_a \times q + c) + t \times (b - d)$
- 5 **Optimal** allocation

Major Limitation

- Unrealistic nature of assumptions 1 and 3 (polycentrism/heterogeneity)
- Lack of temporal dynamics (static cities)
- To be tested against data: the cases of Paris and Lyon

Evolution of Employment Centers in the Los Angeles Metropolitan Region (1995–2005)

	1995	2000	2005
Total number of centers	48	53	53
Total employment in centers (thousands)	3,126.3	3,528.9	3,593.5
Share of employment in centers (%)	41.7	42.5	41.2

Table: *

Source: Giuliano Small, 95%-10K center data, Los Angeles CSA

Evolution of Parisian Urban Structure

Based on the work of Aguiléra and Mignot

- ① **A hierarchical multipolarization**
- ② **Persistent predominance of the center**
- ③ **Complementary secondary centers**
- ④ **Increasing complexity of mobility flows**

- **Employment Data 1968-2020:**
 - Permanent Equipment Database (INSEE)
 - Population Census (INSEE)
- **Geographic Data:**
 - GEOFLA® - Historical Base Map (IGN)

- **Definition of Centers:**

- Notre-Dame de Paris as the central point for Paris
- Basilique Notre-Dame de Fourvière as the central point for Lyon

- **Data Analysis:**

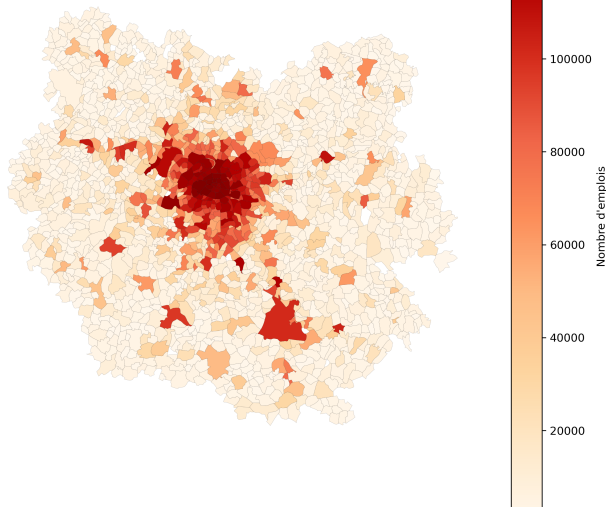
- Calculation of distances from these centers
- Linear regression between distance to center and employment density
- Measurement of job concentration by distance bands

- **Representation of Results:**

- Mapping of job concentrations
- Cumulative distribution curves of jobs
- Graphs of the employment density/distance relationship

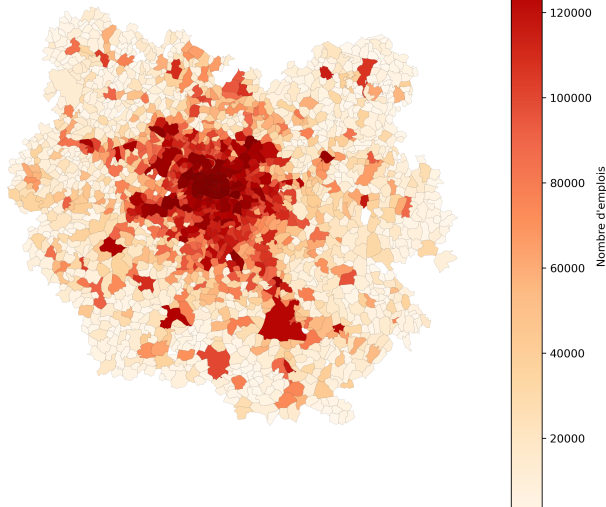
Job Location in Paris in 1968

Nombre d'emplois par commune de l'agglomération parisienne en 1968



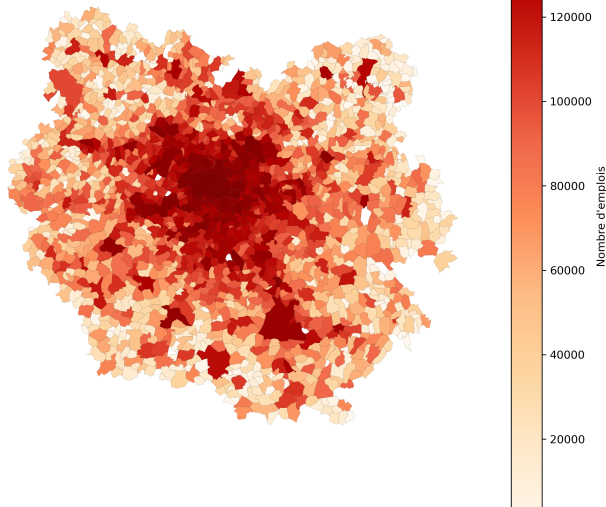
Job Location in Paris in 1999

Nombre d'emplois par commune de l'agglomération parisienne en 1999

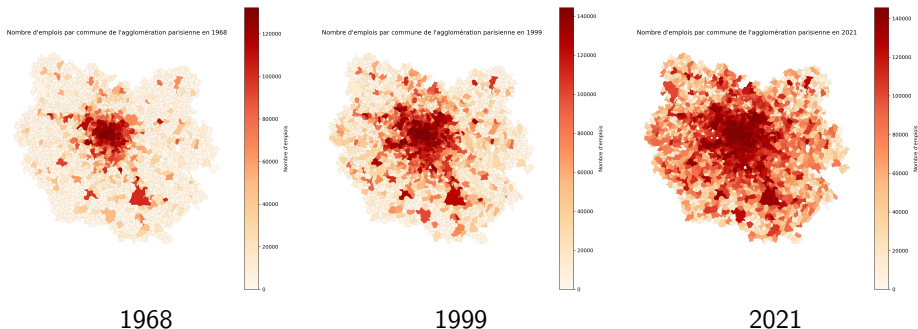


Job Location in Paris in 2021

Nombre d'emplois par commune de l'agglomération parisienne en 2021

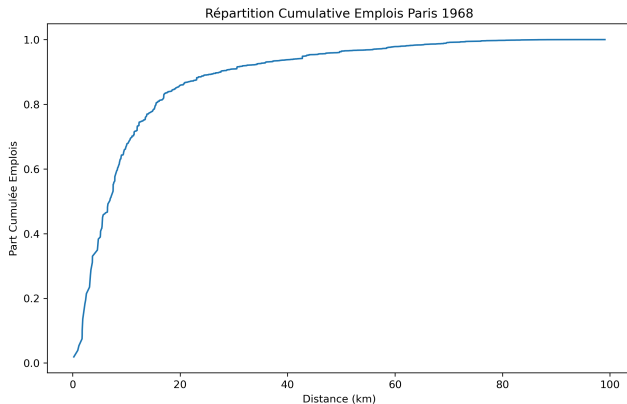


Comparison of Job Distributions in Paris

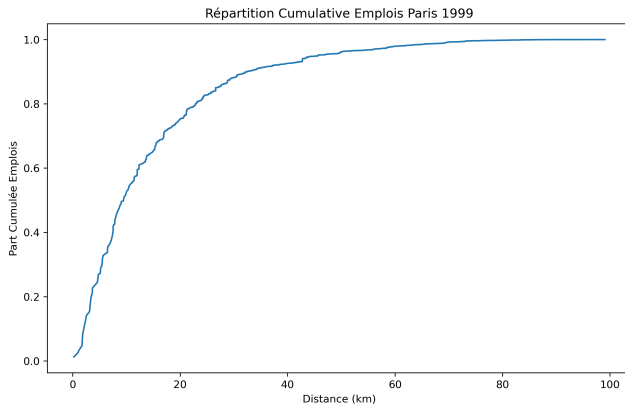


- Evolution of the structure: **monocentric** (1968) → **hierarchical polycentrism** (2021)
- Persistence of strong central concentration but emergence of secondary centers

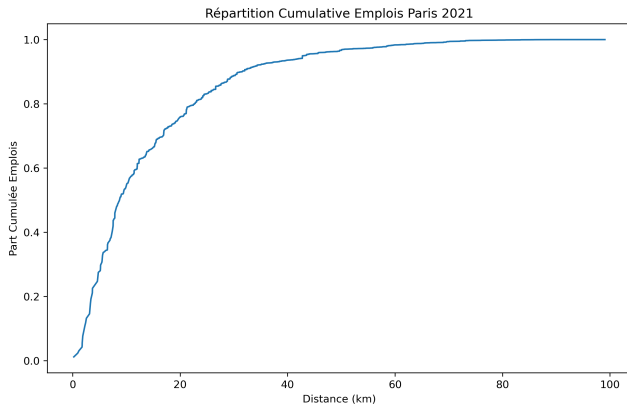
Job Concentration in Île-de-France 1968



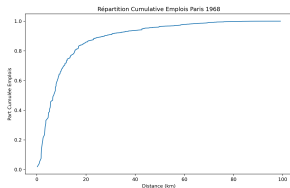
Job Concentration in Île-de-France 1999



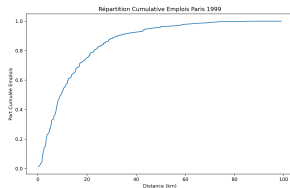
Job Concentration in Île-de-France 2021



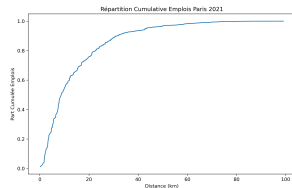
Evolution of Job Concentration in Île-de-France (1968-2021)



1968



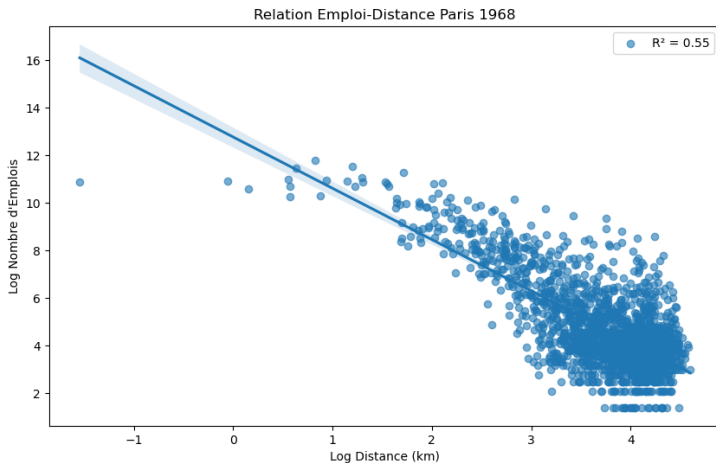
1999



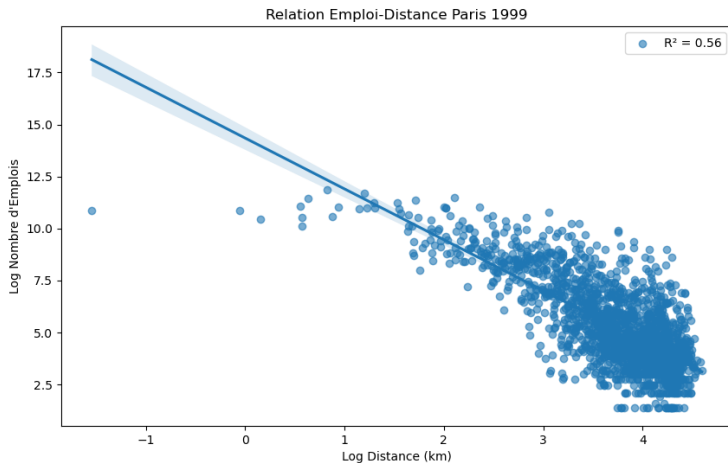
2021

- **Progressive dilution** of centrality over the decades
- Share of jobs within a 10 km radius: **67%** (1968) → **54%** (2021)
- Emergence of **secondary centers**

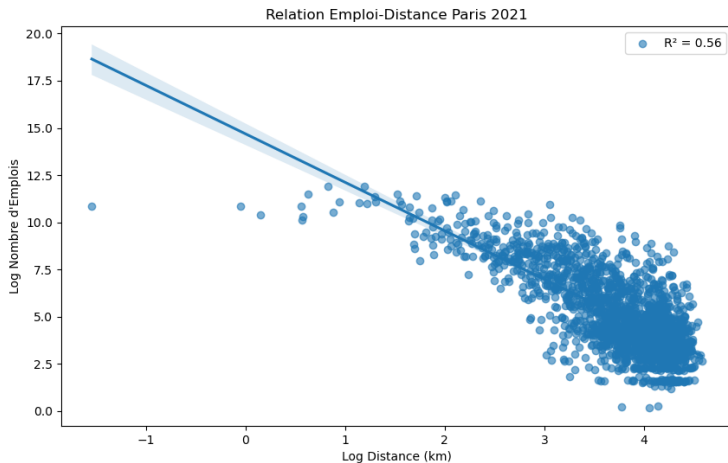
Employment Density and Distance in Île-de-France 1968



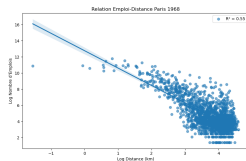
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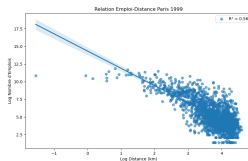
Employment Density and Distance in Île-de-France 2021



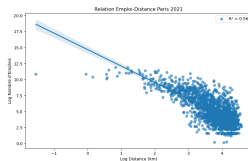
Evolution of the Employment Density/Distance Relationship in Île-de-France (1968-2021)



1968



1999

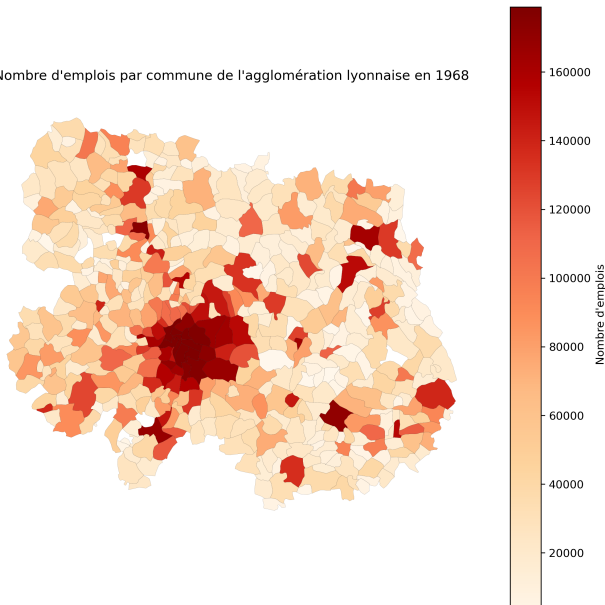


2021

- Employment density gradient: **steep gradient** (1968) → **gentler gradient** (2021)
- Appearance of **secondary peaks** (1999-2021) corresponding to new centers
- **Distorting** effect of transport infrastructure on the density gradient

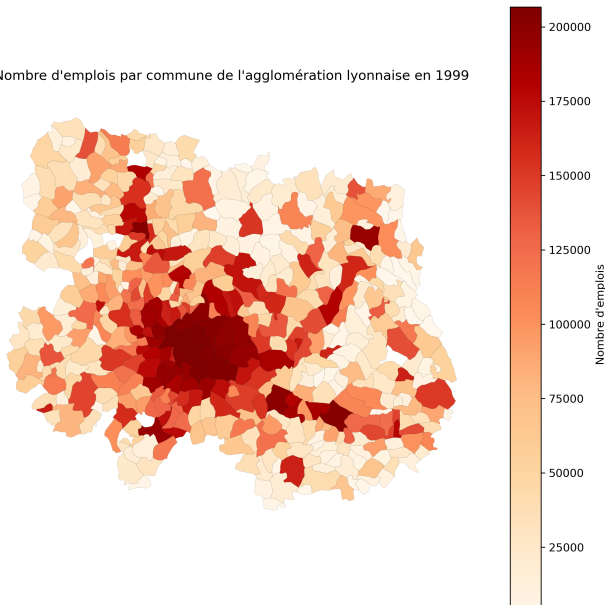
Job Location in the Lyon Agglomeration 1968

Nombre d'emplois par commune de l'agglomération lyonnaise en 1968



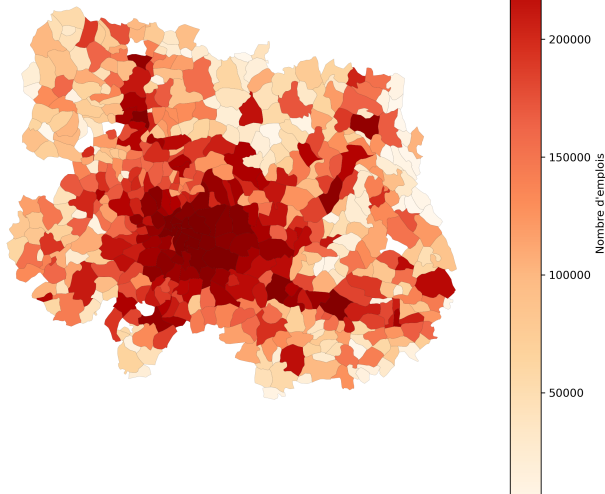
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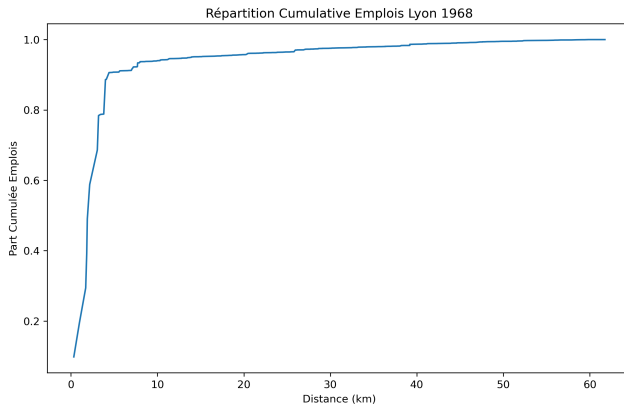


Job Location in the Lyon Agglomeration 2021

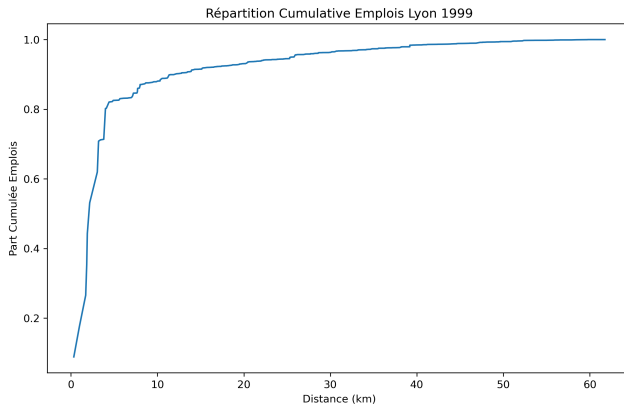
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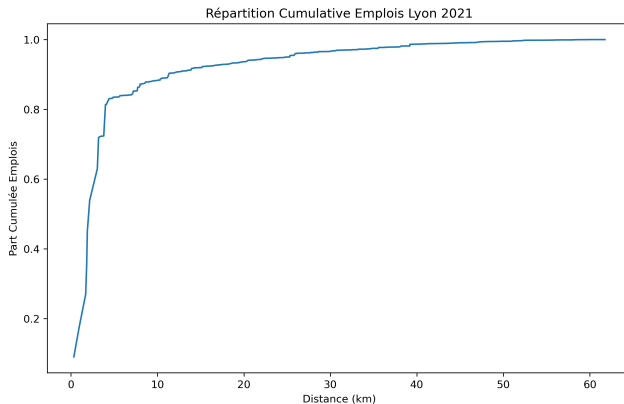
Job Concentration in the Lyon Agglomeration 1968



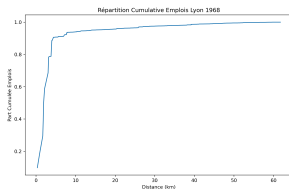
Job Concentration in the Lyon Agglomeration 1999



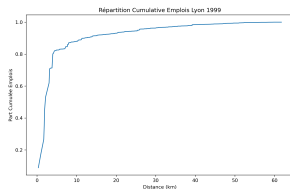
Job Concentration in the Lyon Agglomeration 2021



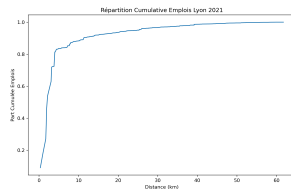
Evolution of Job Concentration in Lyon (1968-2021)



1968



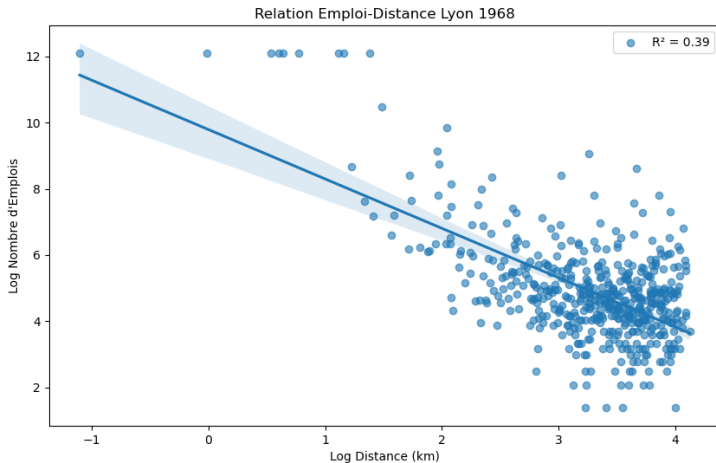
1999



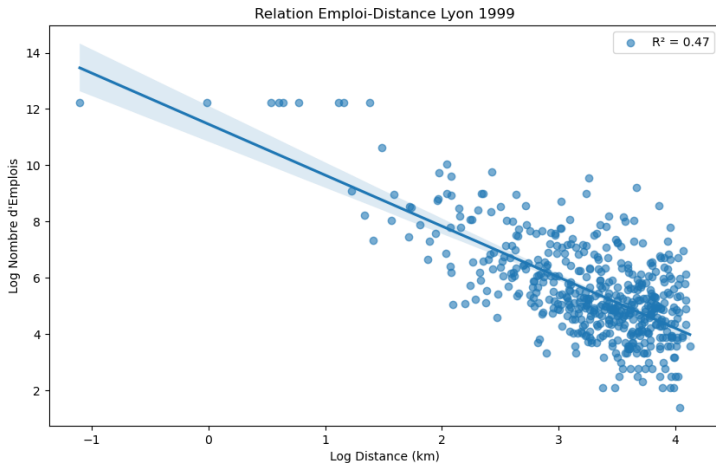
2021

- Evolution from a **strongly monocentric** structure (1968) towards **moderate polycentrism** (2021)
- Persistence of strong concentration: **88%** of jobs remain within a 10 km radius (2021) compared to **94%** in 1968
- Limited emergence of secondary centers (Part-Dieu, Gerland)

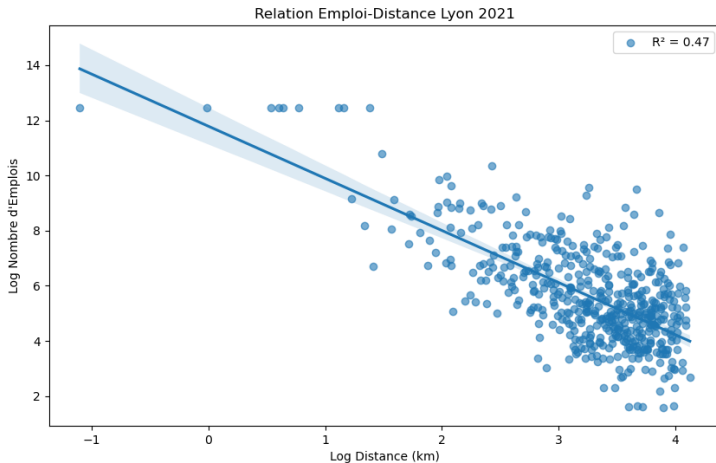
Employment Density and Distance in the Lyon Agglomeration 1968



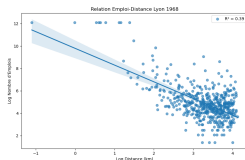
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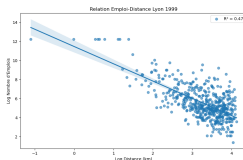
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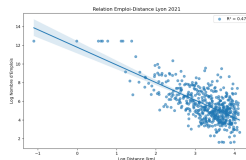
Evolution of the Employment Density/Distance Relationship in Lyon (1968-2021)



1968



1999



2021

- Persistence of a **strong gradient** from the center (1968-2021)
- Less gradient distortion compared to Paris
- Structure still largely **monocentric** despite the appearance of secondary centers

Conclusion

Comparative Synthesis

- **Paris:** **Clear** evolution from a monocentric model (1968) towards **hierarchical polycentrism** (2021)
 - Strong secondary centers (La Défense, Roissy, Saclay)
 - Significant decrease in the relative weight of the center (67)
- **Lyon:** Evolution towards **moderate polycentrism** with **strong persistence** of the center
 - Historic center retains marked dominance
 - Some emerging but limited secondary centers (Part-Dieu, Gerland)

Answer to the Research Question

- Paris is **no longer monocentric**: affirmed polycentric structure
- Lyon remains **predominantly monocentric** despite the emergence of some secondary centers

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Questions?

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